



Operations Research

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Exercise Sheet 8

Demo Exercises

Exercise D8.1. *Unimodularity*

- a) Determine for each of the following matrices whether it is totally unimodular. Justify your answer.

(i) $A = \begin{pmatrix} -1 & 0 & -1 \\ 1 & 1 & 0 \\ 1 & 1 & -1 \end{pmatrix}$

(ii) $B = \begin{pmatrix} 1 & 0 & -1 & 0 \\ 1 & -1 & 0 & 0 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & -1 \end{pmatrix}$

(iii) $C = \begin{pmatrix} -1 & 0 & 0 & -1 & 0 & 0 \\ -1 & -1 & 0 & 0 & -1 & 0 \\ 0 & -1 & -1 & 0 & 0 & -1 \end{pmatrix}$

- b) Determine all possible assignments for x and y such that D is totally unimodular. Justify your answer.

$$D = \begin{pmatrix} x & 1 & -1 \\ 0 & 1 & 0 \\ -1 & y & 1 \end{pmatrix}$$

Exercise D8.2. *DGS Algorithm*

Consider the valuation matrix stated below. Let $p^0 = (0, 0, 0, 0)$ be the initial price vector.

$$\begin{array}{l} A \quad B \quad C \quad D \\ \text{Bidder 1} \begin{pmatrix} 9 & 7 & 4 & 2 \end{pmatrix} \\ \text{Bidder 2} \begin{pmatrix} 6 & 4 & 3 & 3 \end{pmatrix} \\ \text{Bidder 3} \begin{pmatrix} 3 & 1 & 7 & 2 \end{pmatrix} \\ \text{Bidder 4} \begin{pmatrix} 2 & 4 & 6 & 1 \end{pmatrix} \end{array}$$

Compute an optimal allocation using the DGS algorithm and state a competitive equilibrium price vector for all bidders. At every step, state all overdemanded sets of items. If multiple occur at the same step, pick one at random to proceed to the next step.

Exercise D8.3. *Matroids*

First, we define the concept of an independence system. Compare this particularly with the definition of a matroid from the lecture.

Let E be a finite set and $\mathcal{I}$ be a non-empty collection of subsets of E , i.e., $\mathcal{I} \subset \mathcal{P}(E)$. Then $(E, \mathcal{I})$ is an **independence system** if and only if the following hold:

- 1) $\emptyset \in \mathcal{I}$,
 - 2) if $B \in \mathcal{I}$ and $A \subset B$, then $A \in \mathcal{I}$.
- a) Given the sets $E_1 = \{1, 2, 3, 4\}$ and $\mathcal{I}_1 = \{\emptyset, \{1\}, \{2\}, \{3\}, \{4\}, \{1, 2\}, \{1, 3\}, \{1, 4\}, \{2, 3\}, \{1, 2, 3\}, \{1, 2, 4\}\}$.
Prove or disprove: $U_1 = (E_1, \mathcal{I}_1)$ is an independence system.
 - b) Given the sets $E_2 = \{1, 2, 3, 4\}$ and $\mathcal{I}_2 = \{\emptyset, \{1\}, \{2\}, \{3\}, \{4\}, \{1, 2\}, \{1, 3\}, \{1, 4\}, \{2, 3\}, \{2, 4\}, \{3, 4\}, \{1, 2, 4\}, \{2, 3, 4\}\}$.
Prove or disprove: $U_2 = (E_2, \mathcal{I}_2)$ is a matroid.
 - c) Let E_3 be a finite set and let $\mathcal{I}_3 = \{S \subseteq E_3 : |S| \leq k\}$ with $k \in \mathbb{N}$. Show that $U_3 = (E_3, \mathcal{I}_3)$ is a matroid.

Exercise D8.4. *Tips and tricks for network modeling*

A generalization of the Maximum Flow Problem (MFP) is the so-called Minimum Cost Network Flow Problem (MCNFP), which is defined as follows. Given a network $N = (V, E, c, u, b)$ with nodes V , edges E , transportation costs (per unit flow) $c : E \rightarrow \mathbb{R}$, capacities $u : E \rightarrow \mathbb{R}^+$, and demand $b : V \rightarrow \mathbb{R}$, determine a cost-minimal flow $f : E \rightarrow \mathbb{R}$ such that all demands are met, i.e., $\sum_i f(i, j) + b_j = \sum_i f(j, i)$ for all $j \in V$. A positive demand corresponds to a production, and a negative demand corresponds to a consumption.

- a) Show that you can convert any MaxFlow Problem (MFP) into a Minimum Cost Network Flow Problem (MCNFP). To do this, suitably extend the network of the MFP with transportation costs so that an (optimal) solution of the MCNFP corresponds one-to-one to an (optimal) solution of the MFP.

Remark: This procedure is called the *reduction* of MFP to MCNFP. You have shown that MCNFP is at least as difficult as MFP because you can solve any MFP with an MCNFP algorithm and a small additional modeling effort.

- b) Given is a network flow problem with network N_1 where there are more than two nodes with a non-zero demand. Provide a transformed network N_2 that has only one source s and one sink t and whose cost-optimal flows correspond to the cost-optimal flows in the original network up to transformation. (Thus, a reduction from 'MCNFP with multiple sources and sinks' to 'MCNFP with only one source and one sink' is sought.)
- c) Transportation costs can also be negative. On such edges, negative costs result in a profit, for example, in the form of value creation. Reduce MCNFP to 'MCNFP without

negative transportation costs'. It is assumed that an edge with negative costs has a finite capacity.

- d) Given is a network-like model (nodes, edges, capacities) with additional capacity constraints for the flow passing through a *node*. Transform this into an equivalent network where capacities are defined only on edges.